Advanced Machine Learning

Lab 5 --- Module 3

Instructions on how to submit your work on Canvas

To convert your Notebook to an HTML file, follow these steps:

- Open the Notebook that you want to convert to an HTML file.
- Ensure that the notebook is saved with the latest changes.
- In the Notebook interface, navigate to File in the menu bar.
- Click on Download as and select HTML (.html) from the dropdown menu. This will initiate the conversion process.
- The Jupyter Notebook will be converted to an HTML file and automatically downloaded to the default download location on your computer.
- Locate the downloaded HTML file on your computer and verify that the conversion was successful.
- Optionally, open the HTML file in a web browser to view and verify the content.
- Upload your HTML file on Canvas.

Make sure to save your Notebook before attempting to convert it to an HTML file.

Exercise 1 (5 points)

For this exercise, the performance of the SVM is tested in the context of a two-class two-dimensional classification task. The data set comprises $N = 100$ points uniformly distributed in the region $[-5, 5] \times [-5, 5]$. For each point ${\boldsymbol x}_n = [x_{n,1}, x_{n,2}]^{\top}$, we compute

$$y_n = 0.05x_{n,1}^3 + 0.05x_{n,1}^2 + 0.05x_{n,1} + 0.05 + \epsilon,$$

where ϵ stands for zero mean Gaussian noise of variance $\sigma_{\epsilon}^2 = 4$. The point is assigned to either of the two classes, depending on the value of the noise as well as its position with respect to the graph of the function

$$f(x) = 0.05x^3 + 0.05x^2 + 0.05x + 0.05$$

in the two-dimensional space. That is, if $x_{n,2} \geq y_n$, the point is assigned to class ω_1; otherwise, it is assigned to class ω_2.

1. Plot the points $[x_{n,1}, x_{n,2}]$ using different colors for each class.
2. Use SVM with the Gaussian RBF kernel for $\gamma = 15$ and set $C = 1$. Plot the classifier and the margin.
3. Repeat step (b) using $C = 1, 0.1$ and $\gamma = 1$. What do you observe?
4. Repeat step (b) using $C = 0.1$ and $\gamma = 15$. What do you observe?
5. Choose your favorite parameters and apply SVM using a polynomial kernel. How do the results compare with SVM Gaussian kernel?